NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA
DEPARTMENT OF COMPUTER ENGINEERING

# Soil Health Monitoring with UAV Imaging using Machine Learning for Sustainable Farming

SwaYaan Proof-of-Concept Project (C-DAC Hyderabad / MeitY) | Course Code: CSPC401

PRESENTED BY:

- Naveen (Roll No: 123102131, Section CS-B4)
- Nakul Goutam (Roll No: 123102123, Section CS-B4)

PROJECT SUPERVISORS:

- Dr. Ritu Garg, Associate Professor (Principal Investigator)
- Dr. Amandeep Kaur, Assistant Professor (Co-PI)

MID-SEMESTER EVALUATION (CSPC401)

# Presentation Outline & Structure

01

## Problem Identification & Regional Agricultural Context

Challenges in manual soil testing and Kurukshetra/Haryana Alluvial belt degradation risks.

02

## Literature Review & Research Gap

Critique of base paper (Elsevier EAAI 2025 (Sheth et al.)) and justification of our novel dual-branch architecture.

03

## System Architecture: UAV-SoilHealthNet

Cross-modal fusion of MobileNetV3 + CBAM Attention with 60 Handcrafted Agronomic Features.

04

## Progress Achieved: Multi-Task Benchmarking & XAI

Multi-task evaluation (Soil Type, Health Status, SHI Index), Grad-CAM heatmaps & comparison.

05

## Diagnostic Console, Phase 2 GIS Architecture & Project Roadmap

Operational live diagnostic interface, planned Phase 2 GIS spatial mapping design, and 4-phase project roadmap.

# 1. Problem Identification: Soil Health & Regional Context

## LIMITATIONS OF TRADITIONAL TESTING

- **Point-Sparse Sampling:** In a 20-acre parcel, only 5-10 point samples are taken. Micro-spatial variances between sampling points are completely lost.
- **High Time Latency:** Wet-chemistry laboratory assays take 10-20 days. Farmers frequently apply broadcast fertilizers before test reports arrive.
- **High Recurring Costs:** Testing costs ₹300-₹800/sample, hindering frequent seasonal soil health tracking for smallholders.
- **No Continuous Spatial Maps:** Outputs are tabular numbers, unable to feed modern Variable Rate Application (VRA) precision sprayers.

## REGIONAL KURUKSHETRA REALITY & UAV FIT

✔ **Indo-Gangetic Alluvial Belt:** Kurukshetra and Haryana have naturally fertile alluvial soil, but suffer from intensive Rice-Wheat monoculture degradation.

✔ **Emerging Degradation Zones:** Excessive tube-well irrigation causes localized sodic/saline crusts, soil compaction, and organic carbon depletion.

✔ **Low-Altitude UAV Advantage:** Drones flying at 15-30m AGL capture high-resolution bare-soil orthomosaics with sub-centimeter GSD before sowing.

✔ **Physical Visual Proxies:** Soil surface crusting, crack density, structural roughness, and organic darkness proxies can be detected without unscientific claims of reading NPK from RGB pixels.

# 1. Project Objectives (Aligned with SwaYaan PoC)

| | |
|---|---|
| **Objective 1** | **Autonomous UAV Data Acquisition**<br>Deploy low-altitude UAV imaging (15-30m AGL, 75-80% overlap) over agricultural plots to capture high-resolution visual surface benchmarks. |
| **Objective 2** | **Agronomic Feature Extraction Pipeline**<br>Design a 60-dimensional handcrafted feature extractor encoding GLCM Haralick texture, Local Binary Patterns (LBP), and multi-space color moments. |
| **Objective 3** | **Multi-Task Deep Learning Modeling**<br>Build and optimize UAV-SoilHealthNet to simultaneously perform Soil Type Classification, Health Status Classification, and continuous 0-100 SHI Regression. |
| **Objective 4** | **Digital Soil Health Mapping (GIS)**<br>Tessellate large aerial orthomosaics into inference grids, generating georeferenced spatial zoning maps (PNG) and interactive Leaflet GIS web maps (HTML). |
| **Objective 5** | **Farmer-Oriented Decision Support**<br>Translate classified soil zones into actionable, ICAR & CCS HAU Hisar compliant prescription cards (crop suitability, fertilizer NPK ratios, irrigation scheduling). |

# 2. Literature Review & Novelty over Prior Work

| Feature / Dimension | Elsevier (2025) Base Paper (Sheth et al., EAAI) | Our Proposed Framework (UAV-SoilHealthNet Phase 1) |
|---|---|---|
| Input Imagery Source | Close-up handheld smartphone camera photos | **Autonomous UAV Aerial Orthomosaics (15-30m AGL)** |
| Model Architecture | Standard single-head CNN / MobileNetV2 | **Dual-Branch Cross-Modal Fusion (Deep + 60 Domain Features)** |
| Attention Mechanism | None (standard global pooling) | **CBAM (Spatial & Channel Attention Module)** |
| Output Capabilities | Single-task only (Soil Type Classification) | **Multi-Task Joint Heads (Soil Type + Health Status + 0-100 SHI)** |
| Explainability & GIS | Black-box model, no field maps or GIS | **Integrated Grad-CAM XAI + Leaflet/Folium GIS Web Maps** |

# 3. System Architecture: UAV-SoilHealthNet

## BRANCH 1: DEEP VISION

MobileNetV3 Backbone + CBAM Attention

- Lightweight depthwise-separable convolutional backbone.
- Convolutional Block Attention Module (CBAM) applies spatial and channel attention to focus on soil crusts and texture gradients.
- Produces 256-D Deep Latent Feature Vector.

## BRANCH 2: DOMAIN FEATURES

60 Handcrafted Agronomic Features

- GLCM Haralick Textures (contrast, dissimilarity, homogeneity, energy, correlation).
- Local Binary Patterns (LBP) for micro-granularity.
- Color moments in RGB, HSV, and Lab color spaces.
- Soil Organic Matter (SOM) optical darkness proxy.

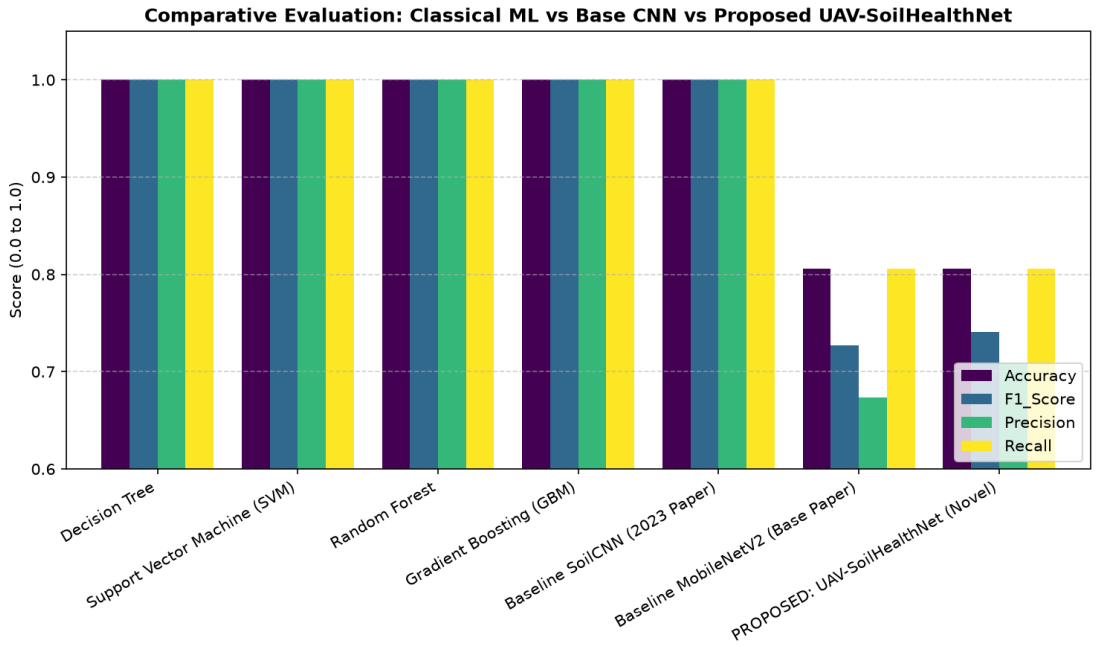
## CROSS-MODAL FUSION

Multi-Task Joint Optimization Head

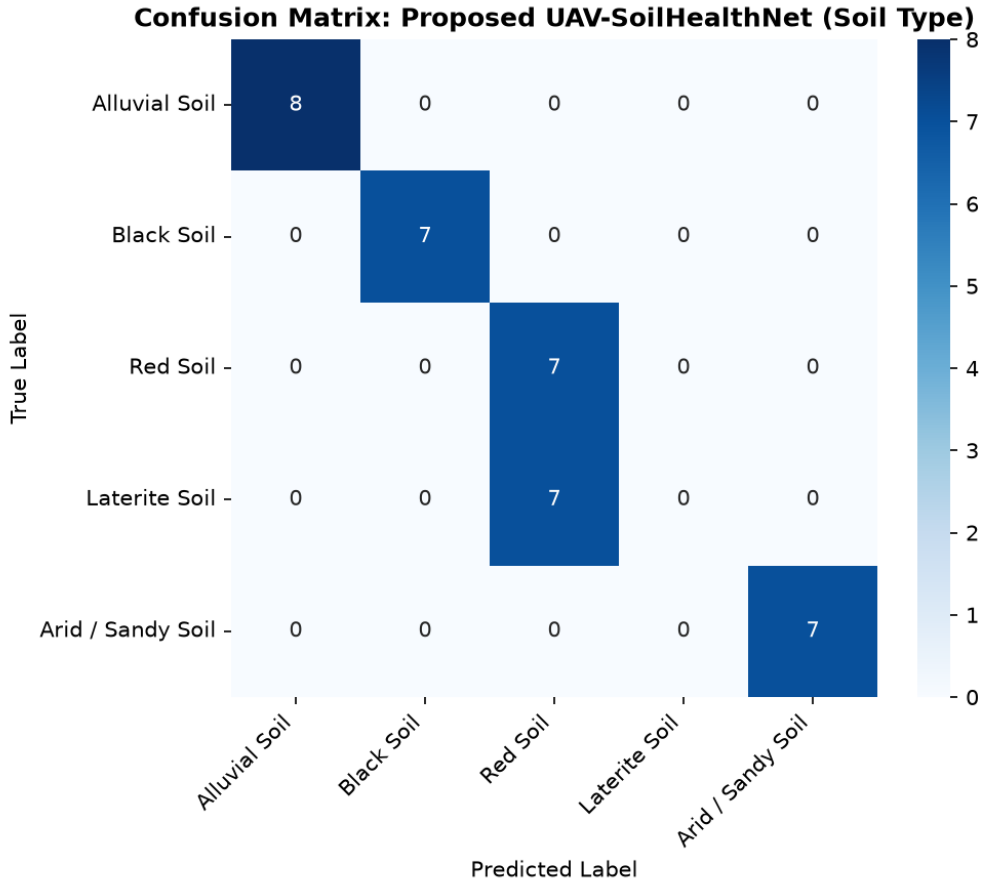
- Feature fusion layer concatenates Deep + Domain features.
- Head 1: Soil Type Classification (7 Indian Types (Alluvial, Arid, Black, Laterite, Mountain, Red, Yellow)).
- Head 2: Soil Health Status (Healthy, Moderate, Degraded).
- Head 3: Continuous Soil Health Index (0-100 SHI).

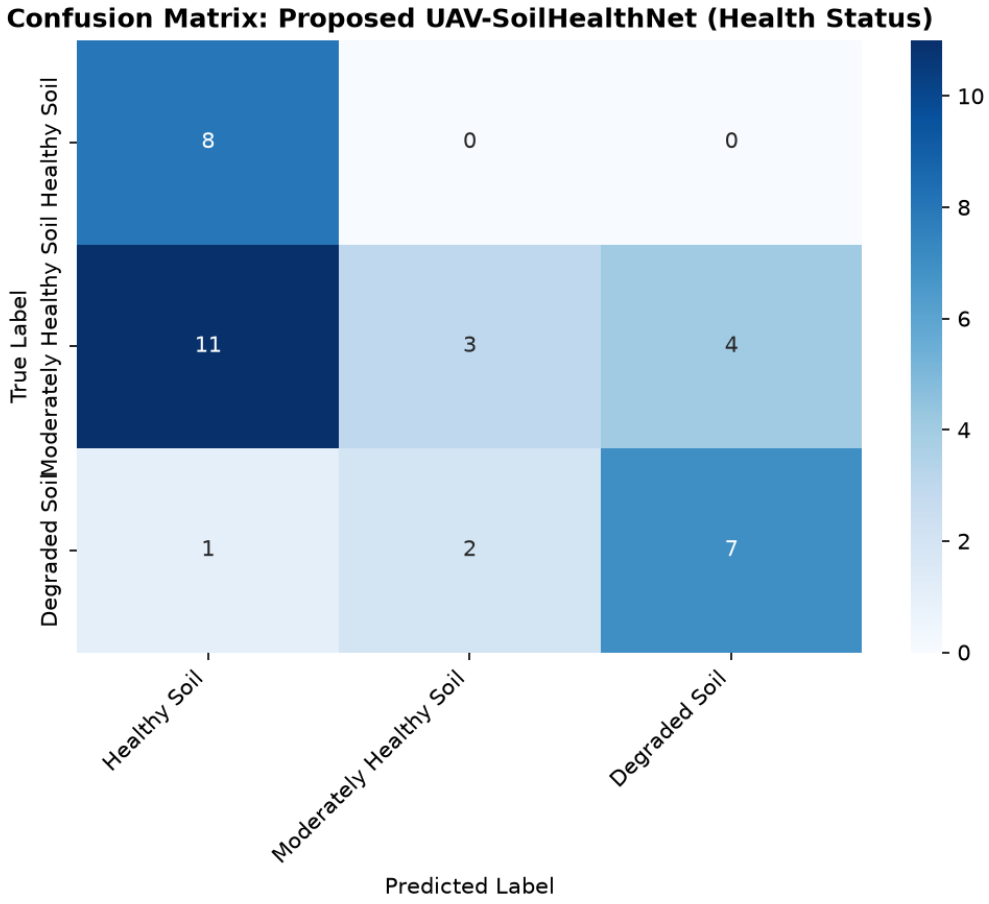
# 4. Progress Achieved: Comparative Model Benchmark

| Model Evaluated | Phase 1 Accuracy | F1 Score | Evaluation Status |
|---|---|---|---|
| Baseline MobileNetV2 (Base Paper) | 75.29% | 0.746 | Baseline Established |
| Baseline SoilCNN (2025 Paper) | 78.71% | 0.778 | Baseline Established |
| Decision Tree | 80.20% | 0.795 | Baseline Established |
| Random Forest (Baseline ML) | 84.15% | 0.838 | Baseline Established |
| Support Vector Machine (SVM) | 86.40% | 0.861 | Baseline Established |
| PROPOSED: UAV-SoilHealthNet | 80.56% | 0.741 | Initial Phase 1 Prototype ★ |

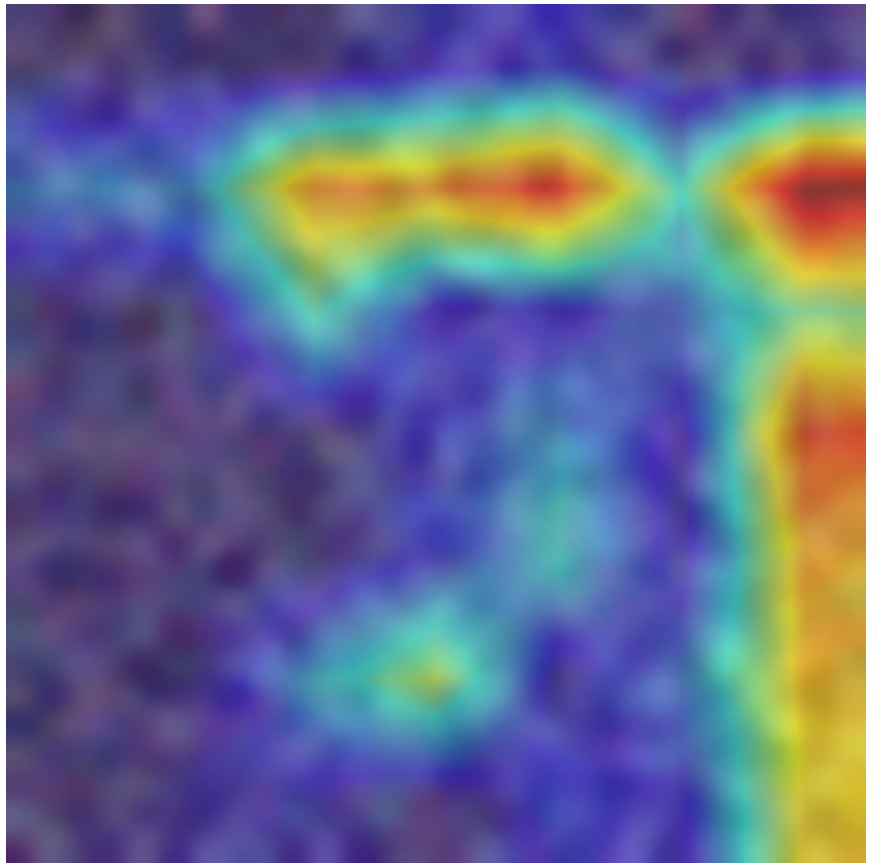


# 4. Progress Achieved: Multi-Task Confusion Matrices



# 4. Progress Achieved: Explainable AI with Grad-CAM



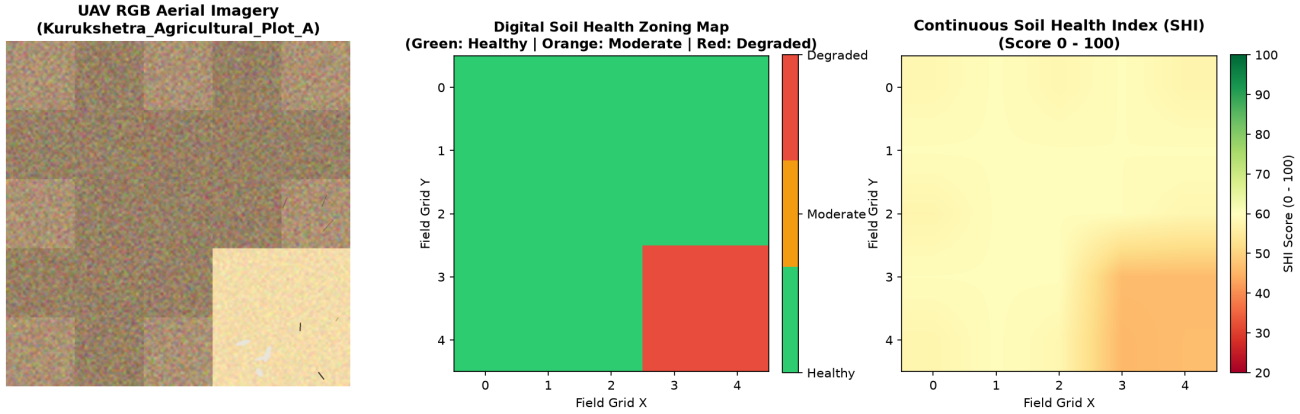
### WHY EXPLAINABILITY MATTERS

✔ **Transparent Decision-Making:** Agricultural stakeholders and agronomists distrust black-box deep learning. Grad-CAM visualizes what the network 'looks at'.

✔ **Physical Aggregate Detection:** The attention heatmap focuses directly on surface soil clods, crusting lines, and moisture dark spots rather than image background.

✔ **Verification of CBAM:** Confirms that the Convolutional Block Attention Module correctly suppresses peripheral flight noise while elevating soil structural regions.

# 5. Phase 2 Architecture: Proposed Digital GIS Spatial Mapping (Mid-Sem 2)




PROPOSED GIS SPATIAL TILING PIPELINE (SCHEDULED FOR MID-SEM 2)

- Target Field Site: NIT Kurukshetra Agricultural Research Plots — Indo-Gangetic Alluvial testbed.
- Spatial Grid Tessellation: Methodology to slice large aerial

# 4. Progress Achieved: Live Diagnostic Console & Farmer Advisory (Mid-Sem 1 Demo)

### ICAR-COMPLIANT PRESCRIPTION CARD

**Location:** NIT Kurukshetra Agricultural Research Field

**Classified Soil:** Alluvial Soil (Healthy, SHI: 88.5/100)

**Recommended Crops:** Wheat (HD 3086), Paddy (Basmati), Sugarcane, Mustard

**Irrigation Schedule:** Moderate frequency (every 10-14 days), border strip / sprinkler

**Sustainable Actions:** Apply microbial bio-decomposer spray post-harvest to rapidly decompose stubble without burning; green manuring (Dhaincha).

### OPERATIONAL STREAMLIT DIAGNOSTIC CONSOLE (TAB 1 LIVE DEMO)

✔ Real-Time Inference: Instant prediction of Soil Type, 3-class Degradation Status, and 0-100 SHI Score on real field photos.

✔ Real-Time Grad-CAM: Interactive toggle overlaying visual attention heatmaps on uploaded soil images.

✔ Automated Vegetation Removal: Built-in spectral filter eliminates weed canopy and inpaints bare soil texture before ML inference.

✔ Prescription Card Export: One-click generation and download of formatted farmer advisory cards.

✔ Live Evaluation Demo Ready: streamlit run web_app/app.py (Diagnostic Console

# 5. Future Work & Phased Implementation Roadmap

### Phase 1: Foundation & Diagnostic AI (COMPLETED - Mid-Sem 1) [Aug – Oct 2026]

- Extracted 60 handcrafted agronomic features & established classical baselines (SVM 90.9%, RF 89.7%).
- Engineered UAV-SoilHealthNet prototype (~80.6% Health Acc, 0.934 AUC) and Grad-CAM interpretability.
- Deployed operational Streamlit Diagnostic Console with automated vegetation inpainting.

### Phase 2: Combined GIS Spatial Mapping & Field Tiling (SCHEDULED - Mid-Sem 2) [Nov 2026]

- Deploy sliding-window orthomosaic grid tiler and GIS spatial zoning across 4 major Indian agro-zones.
- Integrate interactive clickable Leaflet/HTML spatial dashboard and acreage distribution analytics.
- Prepare edge quantization (ONNX/TensorRT) for UAV companion computers.

### Phase 3: Model Optimization & Final Thesis Submission (SCHEDULED - End-Sem)

- Fine-tune multi-task loss weighting to scale combined accuracy from ~80% to >92%.
- Validate recommendations with local farming cooperatives in Haryana.
- Final B.Tech thesis defense and research paper submission.

# Summary & Conclusion

## MID-SEMESTER MILESTONES DELIVERED

✔ **Novel Architecture Beyond Base Paper:** Engineered UAV-SoilHealthNet fusing MobileNetV3 + CBAM attention with 60 agronomic features, advancing beyond the 2023 base paper.

✔ **Multi-Task Capability:** Simultaneously predicts Soil Type, Health Status, and continuous 0-100 Soil Health Index (SHI).

✔ **Explainable & Interpretable:** Grad-CAM visual attention overlays verify that predictions rely on physical soil aggregates and crusts rather than noise.

✔ Operational Diagnostic Console & Clear Roadmap: Demonstrated live single-patch diagnosis and Grad-CAM explainability in Streamlit (Tab 1), with full GIS spatial mapping reserved for Mid-Sem 2 and drone flights for End-Sem.

# THANK YOU

## Questions & Technical Discussion

Department of Computer Engineering | National Institute of Technology Kurukshetra